Multiple regression analysis: data scaling, functional forms, selection of regressors, prediction

Econometrics. Lecture 4

Content

1. Prediction
2. Transformations of variables
3. Selection of regressors

1. Prediction

Prediction.
EViews example (1)

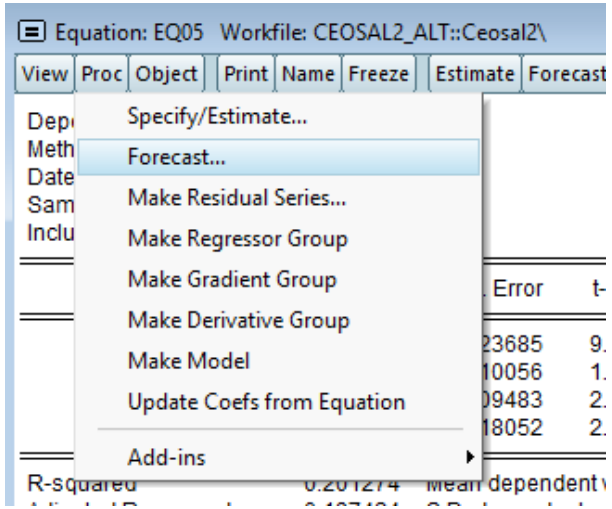
$$\widehat{salary} = 613.44 + 0.019sales + 0.023mktval + 12.703ceoten$$

$$E(salary|sales = 5000, mktval = 10000, ceoten = 10)$$

Dependent Variable: SALARY
Method: Least Squares
Date: 01/02/26 Time: 12:45
Sample: 1 177
Included observations: 177

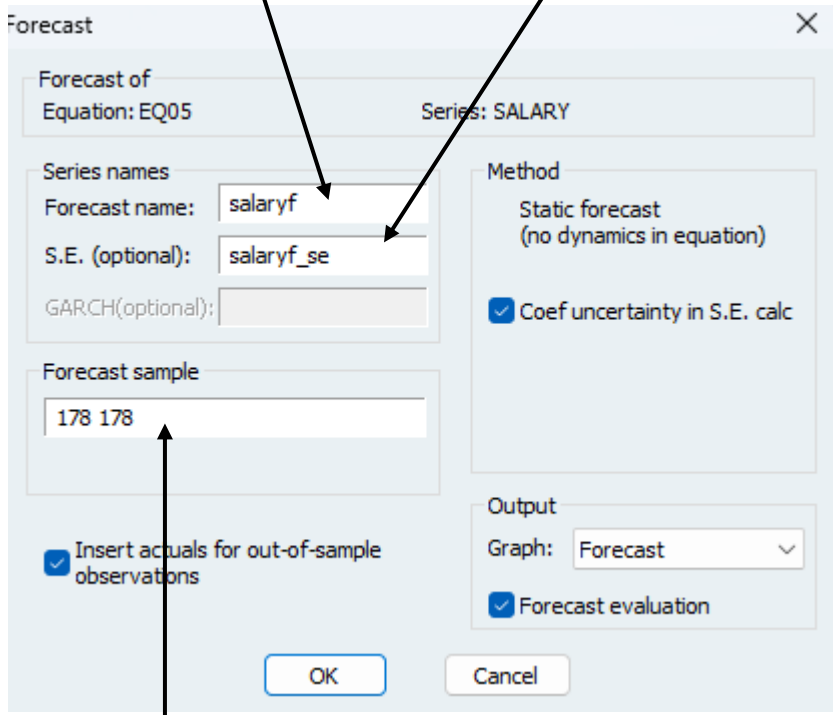
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	613.4361	65.23685	9.403214	0.0000
SALES	0.019019	0.010056	1.891290	0.0603
MKTVAL	0.023400	0.009483	2.467719	0.0146
CEOTEN	12.70337	5.618052	2.261169	0.0250

R-squared	0.201274	Mean dependent var	865.8644
Adjusted R-squared	0.187424	S.D. dependent var	587.5893
S.E. of regression	529.6707	Akaike info criterion	15.40473
Sum squared resid	48535332	Schwarz criterion	15.47651
Log likelihood	-1359.318	Hannan-Quinn criter.	15.43384
F-statistic	14.53168	Durbin-Watson stat	2.166043
Prob(F-statistic)	0.000000		



Specify the name for the forecasted variable

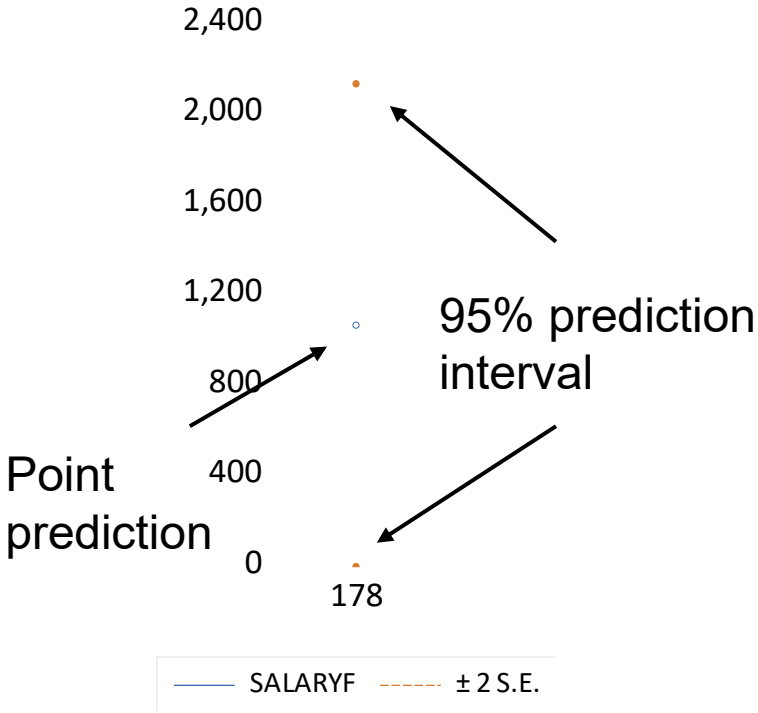
Specify the name for the standard error estimates



salary is a CEO salary, *mktval* is market value of the firm, *ceoten* is years as CEO with the current company.

Specify forecasting range (we add a new observation *I* = 178 and add the values for *sales*, *mktval* and *ceoten*)

Prediction.
EViews example (2)



Series: SALARYF Workfile: CEOSAL2_ALT::Ceosal2\

View | Proc | Object | Properties | Print | Name | Freeze | Default | Sort | Edit+/- | Smpl+/- | Adjust+/-

163	713.0000
164	1350.000
165	1268.000
166	465.0000
167	693.0000
168	369.0000
169	381.0000
170	467.0000
171	559.0000
172	218.0000
173	264.0000
174	185.0000
175	387.0000
176	2220.000
177	445.0000
178	1069.568

Predicting *y* when the dependent variable is in logs (1)

Given the OLS estimators, we know how to predict log*y* for any value of the independent variables:

$$\widehat{\log y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_k x_k$$

Naïve prediction will systematically ***underestimate*** the expected value of *y*.

$$\hat{y} = \exp(\widehat{\log y})$$

Because the model is in logs, we must correct for the fact that variability in log-errors increases the expected value of y. EViews does not do this automatically.

Predicting *y* when the dependent variable is in logs (2)

The consistent prediction of *y* (when *u* is normally distributed)

$$\hat{y} = \exp(\hat{\sigma}^2/2)\exp(\widehat{\log y})$$ where $\hat{\sigma}$ is the standard error of the regression

The consistent prediction of *y* (without normality assumption)

$$\hat{y} = \underbrace{\frac{1}{n}\sum_{i=1}^{n}\exp(\hat{u}_i)}_{\text{A smearing estimate}}\exp(\widehat{\log y})$$

Predicting *y* when the dependent variable is in logs (3)

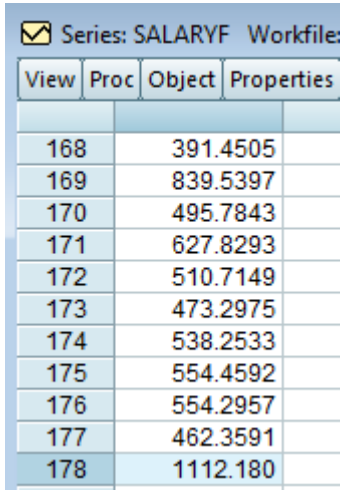
$$\widehat{\log(salary)} = 4.50 + 0.163\log(sales) + 0.109\log(mktval) + 0.012ceoten$$

Dependent Variable: LOG(SALARY)
Method: Least Squares
Date: 01/02/26 Time: 16:21
Sample: 1 177
Included observations: 177

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	4.503795	0.257234	17.50852	0.0000
LOG(SALES)	0.162854	0.039242	4.149993	0.0001
LOG(MKTVAL)	0.109243	0.049595	2.202717	0.0289
CEOTEN	0.011705	0.005326	2.197760	0.0293

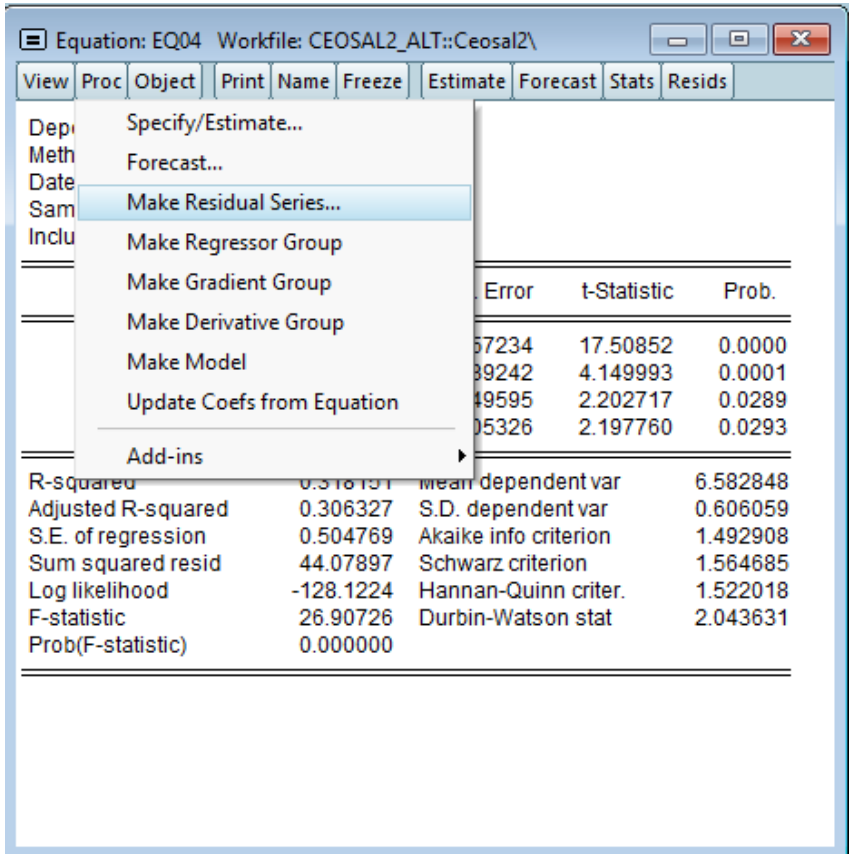
R-squared	0.318151	Mean dependent var	6.582848
Adjusted R-squared	0.306327	S.D. dependent var	0.606059
S.E. of regression	0.504769	Akaike info criterion	1.492908
Sum squared resid	44.07897	Schwarz criterion	1.564685
Log likelihood	-128.1224	Hannan-Quinn criter.	1.522018
F-statistic	26.90726	Durbin-Watson stat	2.043631
Prob(F-statistic)	0.000000		

Naïve prediction



168	391.4505
169	839.5397
170	495.7843
171	627.8293
172	510.7149
173	473.2975
174	538.2533
175	554.4592
176	554.2957
177	462.3591
178	1112.180

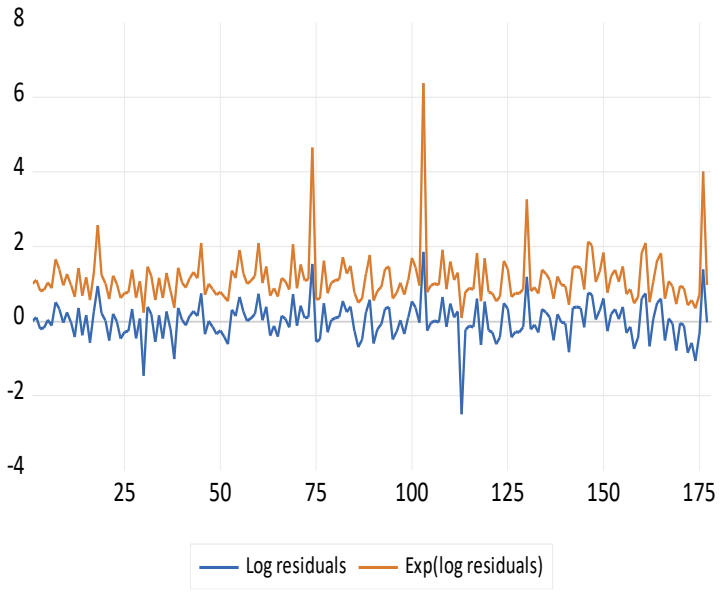
Saving residuals as series



Predicting *y* when the dependent variable is in logs (4)

Calculating the smearing estimate (type in the command window)

scalar smear = @mean(exp(resid01))



Scalar: SMEAR Workfil

View | Proc | Object | Print

1.135661330303012

	Value
SMEAR	1.135661

Calculating the predicted salary (type in the command window)

series yhat = exp(salaryflog) * smear

Series: YHAT Workfile:

View | Proc | Object | Properti

170	563.0430
171	713.0014
172	579.9992
173	537.5057
174	611.2734
175	629.6779
176	629.4922
177	525.0833
178	1263.059



2. Transformations of variables

Interpreting parameters

For a sample of 506 communities in the Boston area, we estimate a model relating median housing price (*price*) in the community to the amount of nitrogen oxide in the air (*nox*) and the average number of rooms in houses in the community (*rooms*).

The population model

$$\log(price) = \beta_0 + \beta_1 \log(nox) + \beta_2 rooms + u$$

β_1 is the **elasticity** of *price* with respect to *nox* (pollution).

β_2 is the change in log(*price*), when $\Delta rooms = 1$;when multiplied by 100, this is the approximate percentage change in price.

Recall that $100*\beta_2$ is the **semi-elasticity** of *price* with respect to *rooms*.

When estimated using the data, we obtain

Dependent Variable: LOG(PRICE)
Method: Least Squares
Date: 01/02/26 Time: 18:43
Sample: 1 506
Included observations: 506

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	9.233738	0.187741	49.18350	0.0000
LOG(NOX)	-0.717673	0.066340	-10.81816	0.0000
ROOMS	0.305918	0.019017	16.08626	0.0000

R-squared	0.513717	Mean dependent var	9.941057
Adjusted R-squared	0.511784	S.D. dependent var	0.409255
S.E. of regression	0.285956	Akaike info criterion	0.339957
Sum squared resid	41.13085	Schwarz criterion	0.365015
Log likelihood	-83.00902	Hannan-Quinn criter.	0.349784
F-statistic	265.6890	Durbin-Watson stat	0.603290
Prob(F-statistic)	0.000000		

When *nox* increases by 1%, *price* falls by 0.718%, holding only rooms fixed.

When *rooms* increases by one, *price* increases by approximately 100*(0.306) = 30.6%.

Approximation is inaccurate for large changes in log(*y*)

Log-level model coefficient interpretation

Using simple algebraic properties of the exponential and logarithmic functions gives the exact percentage change in the predicted *y* as

$$\%\Delta\hat{y} = 100 * \left[\exp\left(\hat{\beta}_2 \Delta x_2\right) - 1\right].$$

When $\Delta x_2 = 1$ $\quad\quad$ $\%\Delta\hat{y} = 100 * \left[\exp\left(\hat{\beta}_2\right) - 1\right].$

Applied to the housing price example

$$\%\Delta\widehat{price} = 100 * [\exp(0.306) - 1] = 35.8\%$$

When $\Delta x_2 = -1$ $\quad\quad$ $\%\Delta\hat{y} = 100 * \left[\exp\left(-\hat{\beta}_2\right) - 1\right].$

$$\%\Delta\widehat{price} = 100 * [\exp(-0.306) - 1] = -26.4\%$$

Considerations when using logarithms

The coefficients have percentage change interpretations. Changing units has no effect on a variable's coefficient when that variable appears in logarithms.

Logs are often used for amounts that are always positive (GDP, wage, population, sales, etc.). They are used less often for variables measured in years (education, age, experience, etc.). Logs are used infrequently for variables that are already percents or proportions (unemployment or interest rates, etc.).

Models with log(y) as the dependent variable often more closely satisfy the CLM assumptions. The model has a better chance of being linear, homoskedasticity is more likely to hold, and normality is often more plausible.

For large changes in an explanatory variable, we can compute a more accurate estimate of the percentage change effect.

Decreasing or increasing effects

Quadratic functions are also used quite often in applied economics to capture decreasing or increasing marginal effects.

For example, consider how wage *y* depends on experience *x*

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + u$$

If we write the estimated equation as

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2$$

then interpreting the partial effect of *x* on *y* we need 2 terms

$$\Delta\hat{y}/\Delta x \approx \hat{\beta}_1 + 2\hat{\beta}_2 x$$

Returns on experience

$$\widehat{wage} = 3.73 + .298\ exper - .0061\ exper^2$$
$$(.35)\quad (.041)\qquad (.0009)$$
$$n = 526,\ R^2 = .093.$$

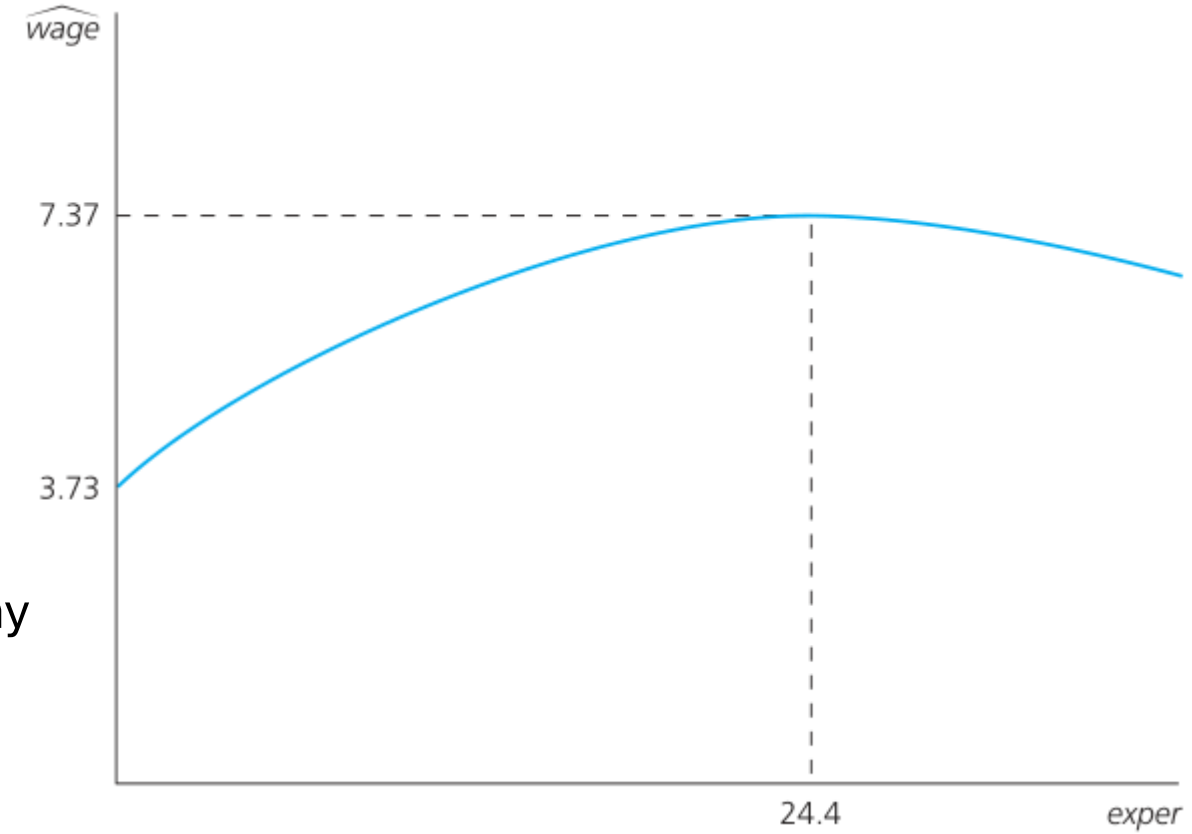
The first year of experience is worth roughly 30¢ per hour ($0.298).

The second year of experience is worth less [about 0.298 – 2*(0.0061)*(1) = 0.286, or 28.6¢].

The general formula for the turning point (x^*) of any quadratic is

$$x^* \approx -\hat{\beta}_1/(2\hat{\beta}_2)$$

Quadratic relationship between wage and experience



Considerations when using quadratics

A quadratic function in an explanatory variable allows for an increasing or decreasing effect.

The turning point of a quadratic is easily calculated, and it should be calculated to see if it makes sense.

Quadratic functions where the coefficients have the opposite sign have a strictly positive turning point; if the signs of the coefficients are the same, the turning point is at a negative value of x.

A seemingly small coefficient on the square of a variable can be practically important in what it implies about a changing slope.

For a model with quadratic in a variable x, the coefficient on x measures the partial effect starting from $x = 0$.

Interaction terms

Sometimes the partial effect of an explanatory variable depends on the magnitude of another explanatory variable.

In the model

$$price = \beta_0 + \beta_1 sqrft + \beta_2 bdrms + \beta_3 sqrft \cdot bdrms + \beta_4 bthrms + u,$$

the partial effect of *bdrms* (number of bedrooms) on house *price* (holding all other variables fixed) is

the effect of *bdrms* on price for a home with zero square feet!

$$\frac{\Delta price}{\Delta bdrms} = \beta_2 + \beta_3 sqrft.$$

If $\beta_3 > 0$, then it implies that an additional bedroom yields a higher increase in housing price for larger houses. There is an **interaction effect** between square footage and number of bedrooms.

Interaction terms. Transformation of variables

Consider a model with two explanatory variables and an interaction.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + u.$$

We can reparameterize the model as

$$y = \alpha_0 + \delta_1 x_1 + \delta_2 x_2 + \beta_3 (x_1 - \mu_1)(x_2 - \mu_2) + u,$$

where μ_1 is the population mean of x_1 and μ_2 is the population mean of x_2.

Now the coefficient on x_2, δ_2, is the partial effect of x_2 on y at the mean value of x_1.

The parameter δ_1 has a similar interpretation.

Interaction terms. EViews example

$$\log(price) = \beta_0 + \beta_1 nearinc + \beta_2 area + \beta_3 baths + \beta_4\left(baths - \overline{baths}\right)(area - \overline{area}) + u$$

House price (*price*), an indicator of the house location (*nearinc*), area of the house (*area*) and the number of bathrooms (*baths*).

Estimated model

Dependent Variable: LOG(PRICE)
Method: Least Squares
Date: 01/03/26 Time: 17:08
Sample: 1 321 IF YEAR=1978
Included observations: 179

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	10.37039	0.072191	143.6517	0.0000
NEARINC	-0.178264	0.042320	-4.212294	0.0000
AREA/1000	0.177305	0.037380	4.743333	0.0000
BATHS	0.210765	0.034279	6.148487	0.0000
(AREA/1000-@MEAN(AREA/1000,"1 321 if ...	0.077750	0.031187	2.493042	0.0136

R-squared	0.599203	Mean dependent var	11.17908
Adjusted R-squared	0.589989	S.D. dependent var	0.366974
S.E. of regression	0.234981	Akaike info criterion	-0.031089
Sum squared resid	9.607597	Schwarz criterion	0.057944
Log likelihood	7.782439	Hannan-Quinn criter.	0.005014
F-statistic	65.03379	Durbin-Watson stat	1.917542
Prob(F-statistic)	0.000000		

Compare with this model

Dependent Variable: LOG(PRICE)
Method: Least Squares
Date: 03/20/26 Time: 12:16
Sample: 1 321 IF YEAR=1978
Included observations: 179

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	10.72910	0.156235	68.67267	0.0000
NEARINC	-0.178264	0.042320	-4.212294	0.0000
AREA/1000	-0.002085	0.087021	-0.023956	0.9809
BATHS	0.055294	0.066652	0.829585	0.4079
(AREA/1000)*(BATHS)	0.077750	0.031187	2.493042	0.0136

R-squared	0.599203	Mean dependent var	11.17908
Adjusted R-squared	0.589989	S.D. dependent var	0.366974
S.E. of regression	0.234981	Akaike info criterion	-0.031089
Sum squared resid	9.607597	Schwarz criterion	0.057944
Log likelihood	7.782439	Hannan-Quinn criter.	0.005014
F-statistic	65.03379	Durbin-Watson stat	1.917542
Prob(F-statistic)	0.000000		

Considerations when using interactions

Interaction terms allow the partial effect of an explanatory variable, say x_1, to depend on the level of another variable, say x_2 – and vice versa.

Interpreting models with interactions can be tricky. The coefficient on x_1, say β_1, measures the partial effect of x_1 on y when $x_2 = 0$, which may be impossible or uninteresting. Centering x_1 and x_2 around interesting values before constructing the interaction term typically leads to an equation that is visually more appealing.

A standard t-test can be used to determine if an interaction term is statistically significant. Computing the partial effects at different values of the explanatory variables can be used to determine the practical importance of interactions.

3. Selection of regressors

Adjusted R-squared

R^2 can never fall when a new independent variable is added to a regression equation.

The primary attractiveness of **adjusted R-squared** ($\bar{R}^2$) is that it imposes a penalty for adding additional independent variables to a model.

$$\bar{R}^2 = 1 - (1 - R^2) * \frac{n-1}{n-k-1}$$

If we add a new independent variable to a regression equation, adjusted R^2 increases if, and only if, the *t*-statistic on the new variable is greater than one in absolute value. This makes $\bar{R}^2$ preferable for choosing between models with different numbers of explanatory variables.

Criteria for model comparison

Akaike Information Criterion

$$AIC = n\ln(SSR) + 2k$$

Schwarz Bayesian Criterion

$$SBC = n\ln(SSR) + k\ln(n)$$

Ideally, we select the model that minimizes all those statistics. However, in general the AIC is one of the most commonly used.

Model comparison

Model 1

Dependent Variable: LOG(SALARY)
Method: Least Squares
Date: 01/03/26 Time: 17:20
Sample: 1 353
Included observations: 353

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	11.34433	0.107645	105.3866	0.0000
YEARS	0.068106	0.012123	5.617899	0.0000
GAMESYR	0.016228	0.001525	10.63947	0.0000
HRUNSYR	0.035863	0.007249	4.947639	0.0000

R-squared	0.623481	Mean dependent var	13.49218
Adjusted R-squared	0.620244	S.D. dependent var	1.182466
S.E. of regression	0.728687	Akaike info criterion	2.216121
Sum squared resid	185.3134	Schwarz criterion	2.259933
Log likelihood	-387.1453	Hannan-Quinn criter.	2.233554
F-statistic	192.6373	Durbin-Watson stat	1.217607
Prob(F-statistic)	0.000000		

Model 2

Dependent Variable: LOG(SALARY)
Method: Least Squares
Date: 01/03/26 Time: 17:22
Sample: 1 353
Included observations: 353

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	11.44738	0.112831	101.4561	0.0000
YEARS	0.069899	0.012061	5.795353	0.0000
GAMESYR	0.011262	0.002139	5.266232	0.0000
RBISYR	0.016782	0.003205	5.236635	0.0000

R-squared	0.626425	Mean dependent var	13.49218
Adjusted R-squared	0.623214	S.D. dependent var	1.182466
S.E. of regression	0.725832	Akaike info criterion	2.208271
Sum squared resid	183.8645	Schwarz criterion	2.252084
Log likelihood	-385.7598	Hannan-Quinn criter.	2.225704
F-statistic	195.0722	Durbin-Watson stat	1.262635
Prob(F-statistic)	0.000000		

Model 3

Dependent Variable: LOG(SALARY)
Method: Least Squares
Date: 01/03/26 Time: 17:24
Sample: 1 353
Included observations: 353

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	11.42672	0.116641	97.96469	0.0000
YEARS	0.069279	0.012102	5.724753	0.0000
GAMESYR	0.012349	0.002636	4.684628	0.0000
HRUNSYR	0.011000	0.015579	0.706045	0.4806
RBISYR	0.012456	0.006915	1.801347	0.0725

R-squared	0.626959	Mean dependent var	13.49218
Adjusted R-squared	0.622671	S.D. dependent var	1.182466
S.E. of regression	0.726354	Akaike info criterion	2.212505
Sum squared resid	183.6015	Schwarz criterion	2.267271
Log likelihood	-385.5072	Hannan-Quinn criter.	2.234297
F-statistic	146.2185	Durbin-Watson stat	1.252131
Prob(F-statistic)	0.000000		